Lower Limb Neurology

Prof Isaac Sorinola

Learning Outcomes

By the end of this lecture, students should be able to:

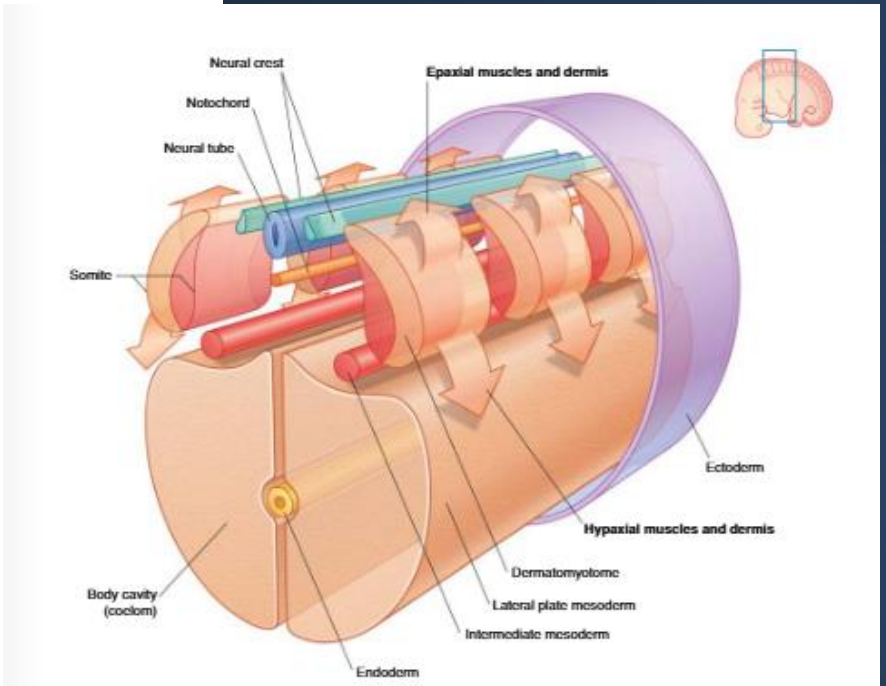
- Explain what is meant by the term dermatome and myotome and their relationship to the development of the limb
- Describe dermatomal and myotomal maps of the lower limb,
- Describe structure and branches of the lumbar and sacral plexus
- Describe the course and distribution of common lower limb nerves such as femoral, obturator, sciatic, common peroneal and tibial nerves.
- Describe cutaneous nerve distribution of the lower limb

Development of Dermatomes & Myotomes

Functionally, the nervous system can be divided into:

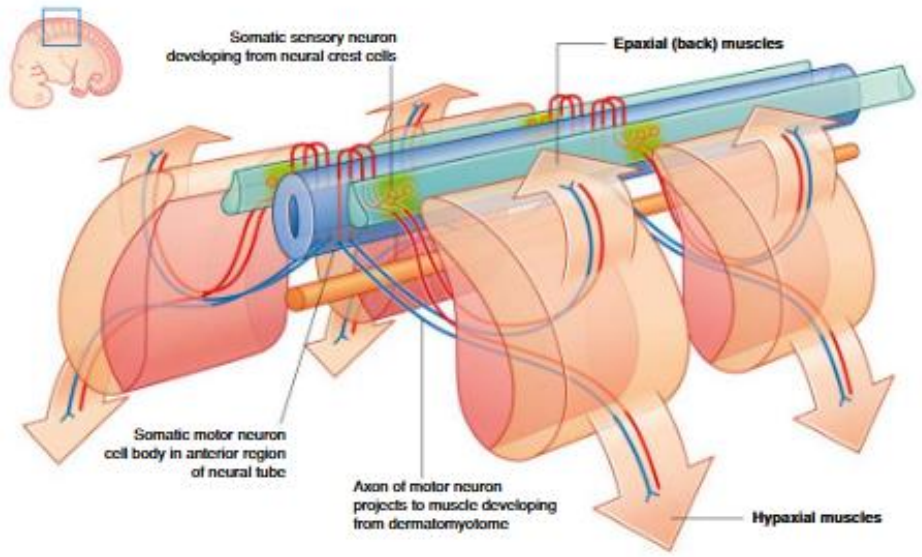
- *Somatic part*
  - Innervates structures – skin and most skeletal muscle
  - Somatic nerves arise segmentally along with developing CNS in association with somites

- *Visceral part*
  - Innervates organ systems in the body and other visceral elements such as smooth muscle and glands, in peripheral regions of the body.



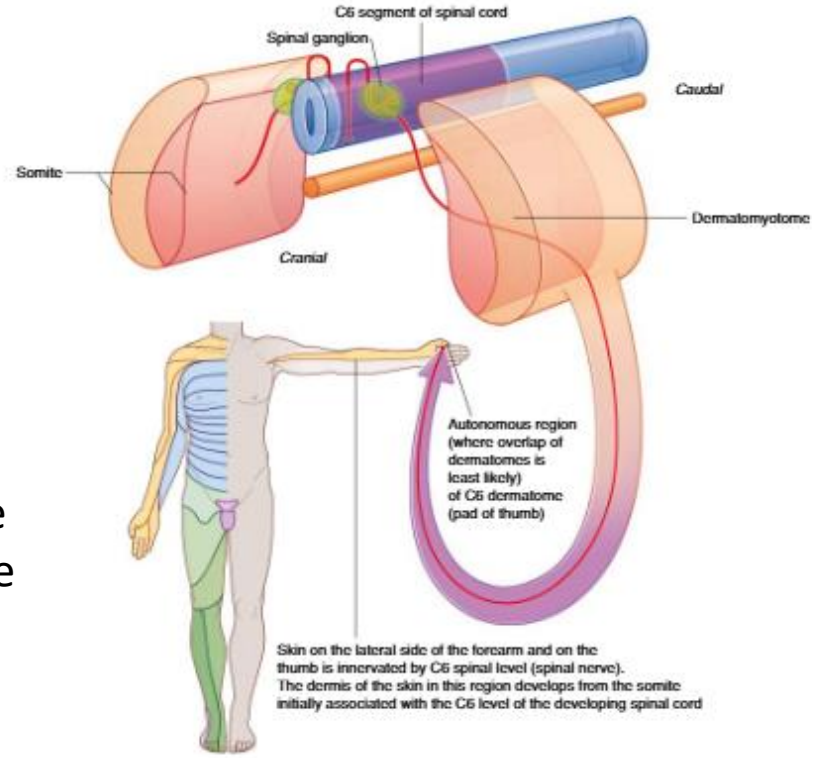
Development of dermatomes and myotomes

Neural crest cells differentiate into neurons on each side of the neural tube and form the sensory neurons



Neurons also develop from within the neural tube, and they form the motor neuron

Because cells from a specific somite develop into the dermis of the skin in a specific location, somatic sensory fibres originally associated with that somite carries information from that region and become part of one spinal nerve



Somatic motor nerves that were originally associated with a specific somite become part of one spinal nerve

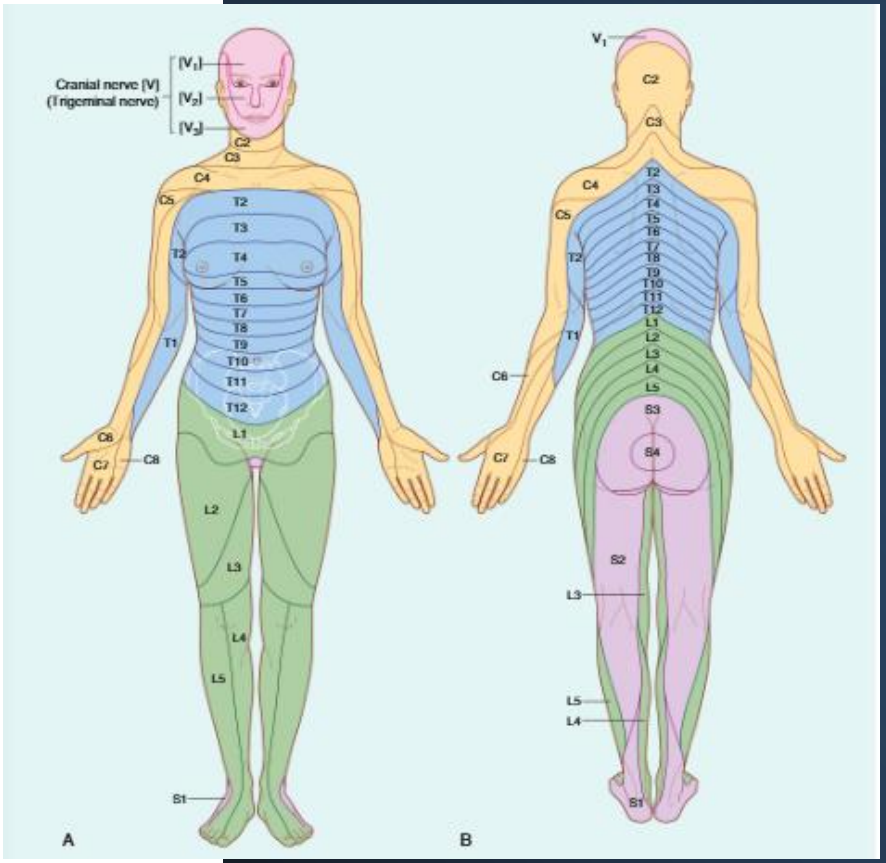
Consequently…

A dermatome *is that area of skin supplied by a single spinal cord level or by a single spinal nerve*

A myotome *is that portion of a skeletal muscle innervated by a single spinal cord level or by a single spinal nerve*

Clinical Application

- *Knowledge of dermatomes and myotomes - essential in neurological examination*
- *Dermatome: light touch or pin prick sensation testing*
- *Myotome: testing joint movements or specific muscle group*



INTERNATIONAL STANDARDS FOR NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY (ISNCSCI)

ASIA American Spinal Injury Association | ISCOS International Spinal Cord Society

Patient Name _____ Date/Time of Exam _____

Examiner Name _____ Signature _____

RIGHT

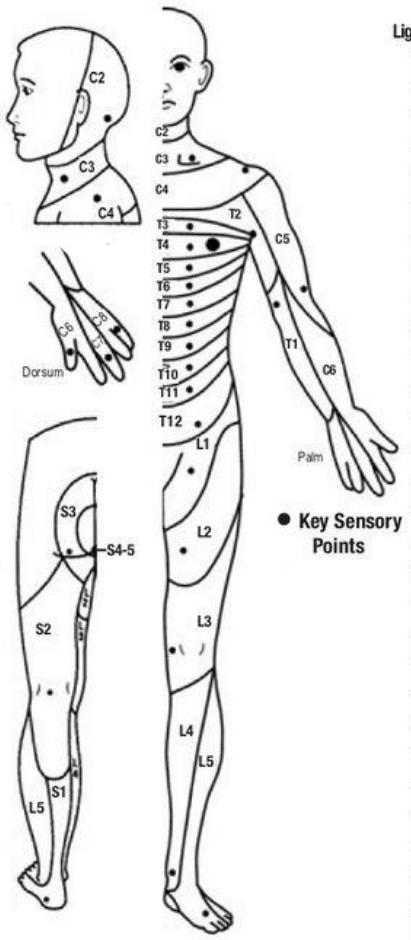
MOTOR KEY MUSCLES		SENSORY KEY SENSORY POINTS Light Touch (LTR)	Pin Prick (PPR)
	C2		
	C3		
	C4		
Elbow flexors	C5		
Wrist extensors	C6		
Elbow extensors	C7		
Finger flexors	C8		
Finger abductors (little finger)	T1		
	T2		
	T3		
	T4		
	T5		
	T6		
	T7		
	T8		
	T9		
	T10		
	T11		
	T12		
	L1		
Hip flexors	L2		
Knee extensors	L3		
Ankle dorsiflexors	L4		
Long toe extensors	L5		
Ankle plantar flexors	S1		
	S2		
	S3		
	S4-5		
RIGHT TOTALS (MAXIMUM)		(50)	(56) (56)

UER (Upper Extremity Right)

LER (Lower Extremity Right)

Comments (Non-key Muscle? Reason for NT? Pain?):

(VAC) Voluntary Anal Contraction (Yes/No) ☐



LEFT

SENSORY KEY SENSORY POINTS Light Touch (LTL)	Pin Prick (PPL)	MOTOR KEY MUSCLES	
		C2	
		C3	
		C4	
		C5	Elbow flexors
		C6	Wrist extensors
		C7	Elbow extensors
		C8	Finger flexors
		T1	Finger abductors (little finger)
		T2	
		T3	
		T4	
		T5	
		T6	
		T7	
		T8	
		T9	
		T10	
		T11	
		T12	
		L1	
		L2	Hip flexors
		L3	Knee extensors
		L4	Ankle dorsiflexors
		L5	Long toe extensors
		S1	Ankle plantar flexors
		S2	
		S3	
		S4-5	
(56)	(56)	(50)	LEFT TOTALS (MAXIMUM)

UEL (Upper Extremity Left)

LEL (Lower Extremity Left)

MOTOR (SCORING ON REVERSE SIDE)

0 = total paralysis
1 = palpable or visible contraction
2 = active movement, gravity eliminated
3 = active movement, against gravity
4 = active movement, against some resistance
5 = active movement, against full resistance
5* = normal corrected for pain/disuse
NT = not testable

SENSORY (SCORING ON REVERSE SIDE)

0 = absent 2 = normal
1= altered NT = not testable

(DAP) Deep Anal Pressure (Yes/No) ☐

MOTOR SUBSCORES

UER ☐ + UEL ☐ = *UEMS TOTAL* ☐ — MAX (25) (25) (50)

LER ☐ + LEL ☐ = *LEMS TOTAL* ☐ — MAX (25) (25) (50)

SENSORY SUBSCORES

LTR ☐ + LTL ☐ = *LT TOTAL* ☐ — MAX (56) (56) (112)

PPR ☐ + PPL ☐ = *PP TOTAL* ☐ — MAX (56) (56) (112)

NEUROLOGICAL LEVELS
Steps 1-5 for classification as on reverse

	R	L
1. SENSORY	☐	☐
2. MOTOR	☐	☐

3. NEUROLOGICAL LEVEL OF INJURY (NLI) ☐

4. COMPLETE OR INCOMPLETE? ☐
Incomplete = Any sensory or motor function in S4-5

5. ASIA IMPAIRMENT SCALE (AIS) ☐

(In complete injuries only)
ZONE OF PARTIAL PRESERVATION
Most caudal level with any innervation

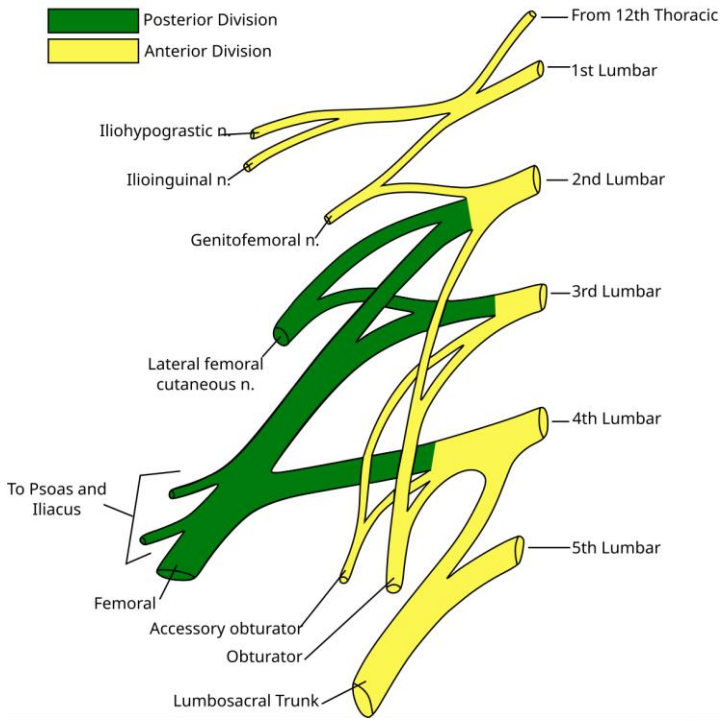
	R	L
SENSORY	☐	☐
MOTOR	☐	☐

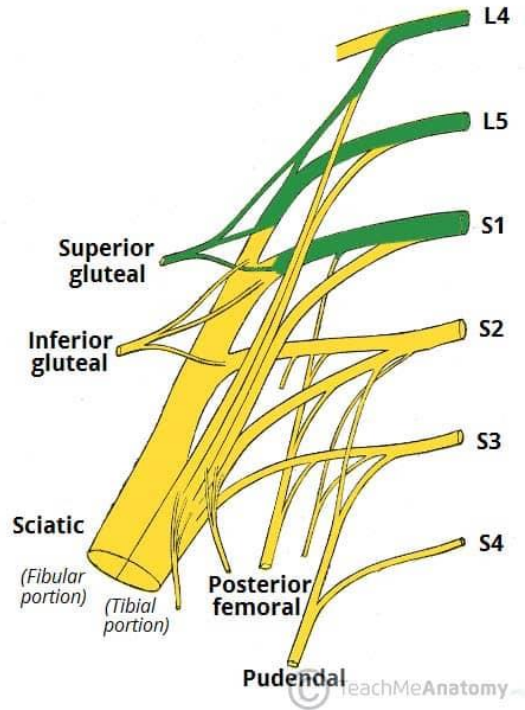
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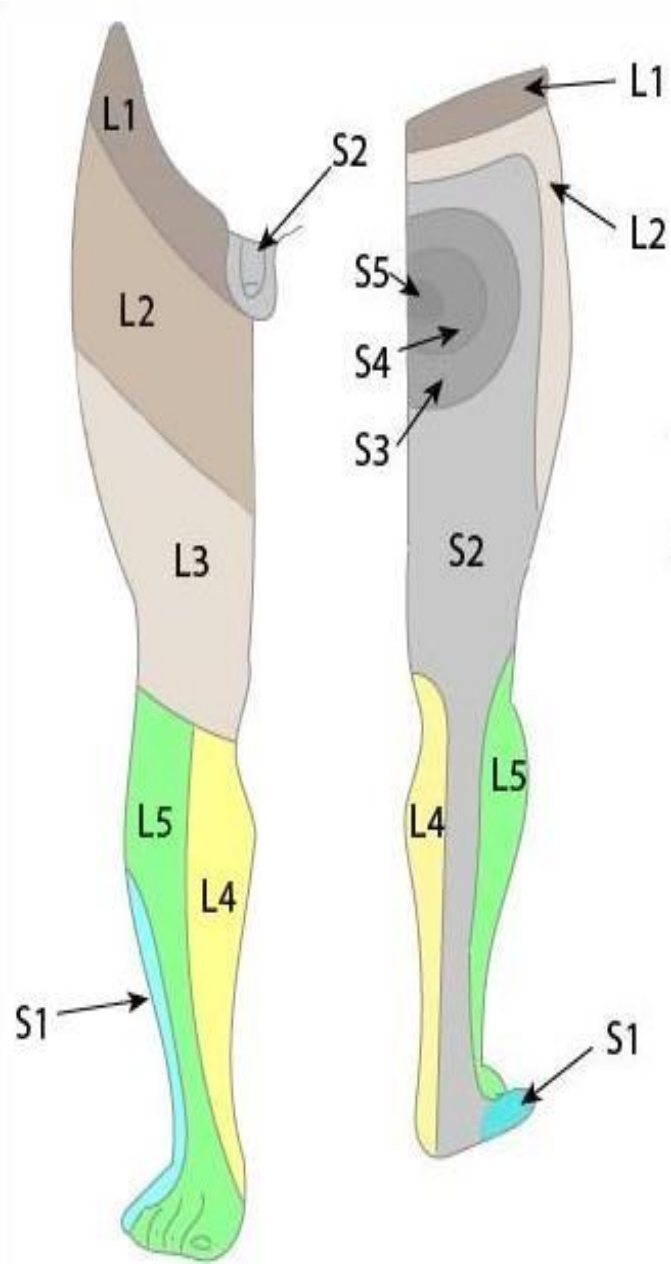
Organisation of the lower limb neurology

- *Lumbar plexus*
  - Anterior rami of **L1 - L3 & most of L4**
- *Sacral Plexus*
  - Anterior rami of **L4 to S5**
- *Nerves entering the lower limb carry fibres from L1 – S3; lower sacral nerves innervate the perineum (S3–S5).*
- *Lumbar and upper sacral nerves are tested clinically in the LL.*

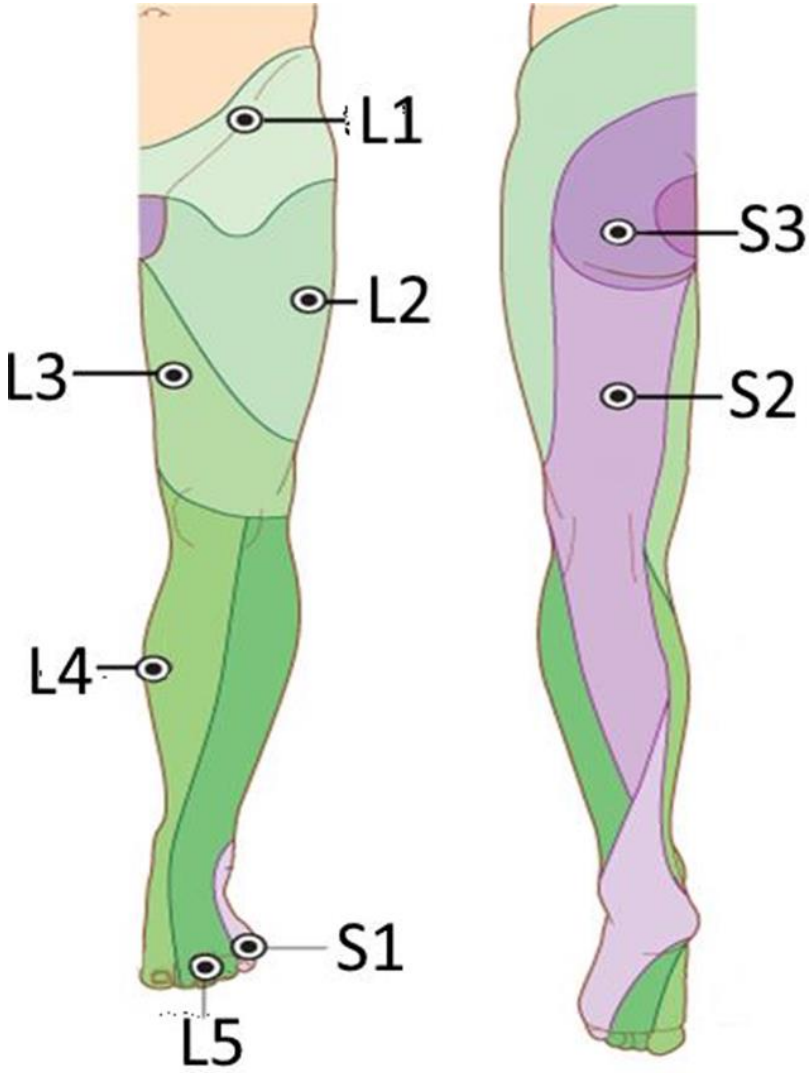




Dermatomes



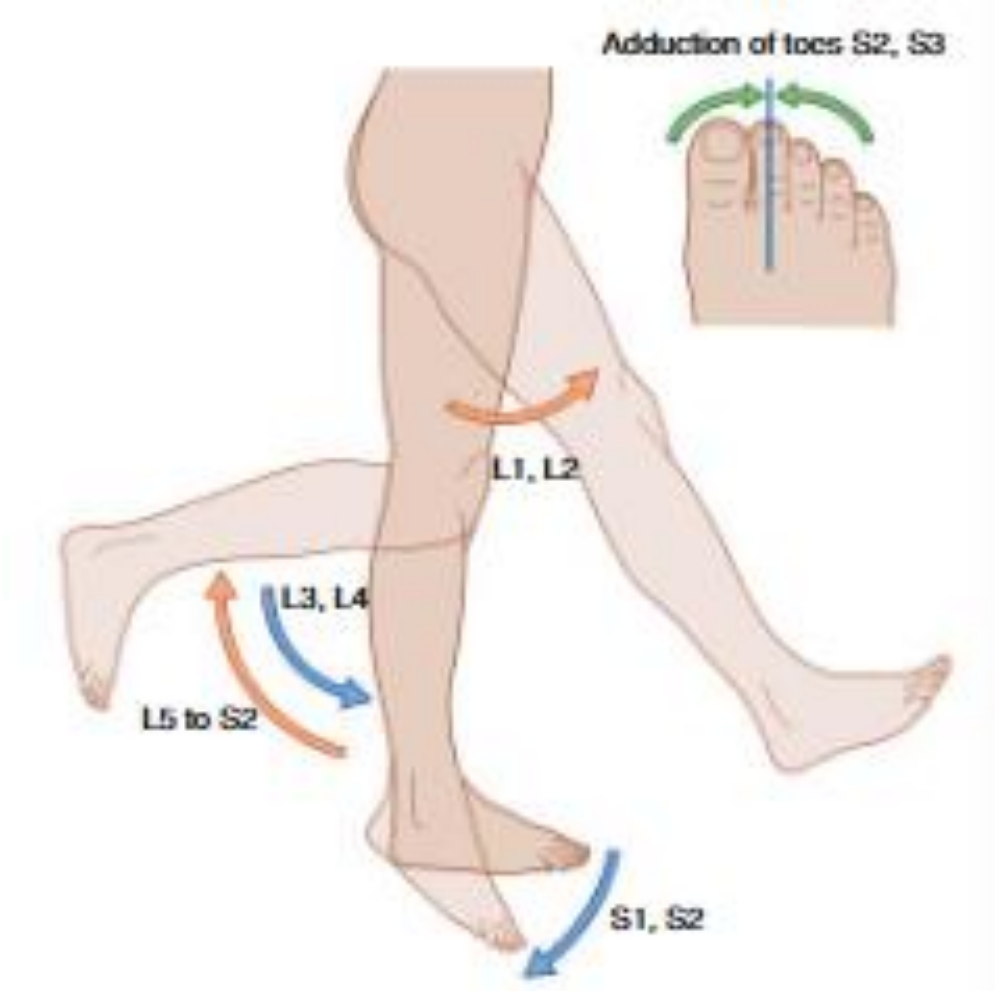
Testing Dermatomes



Nerve root	Autonomous region (minimum overlap)
L1	Over the inguinal ligament
L2	Lateral side of the thigh
L3	Lower medial side of thigh
L4	Medial side of great toe
L5	Medial side of digit 2
S1	Little toe (digit 5)
S2	Back of thigh
S3	Skin over gluteal fold

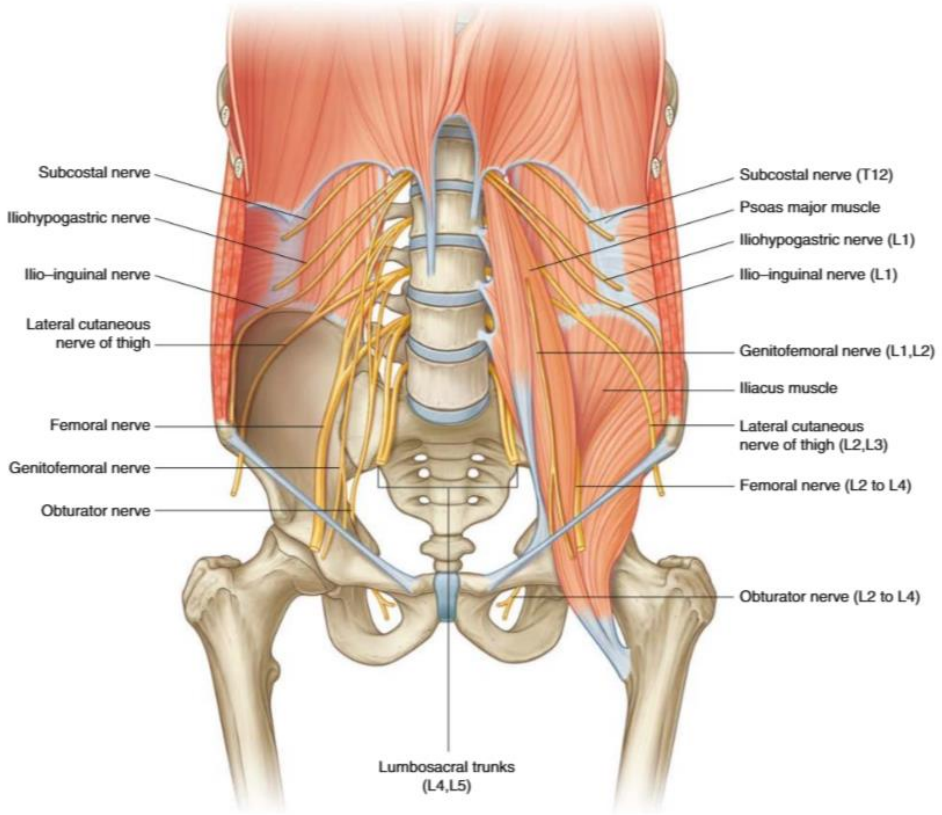
S4 & S5 are tested in the perineum

Lower Limb Myotomes



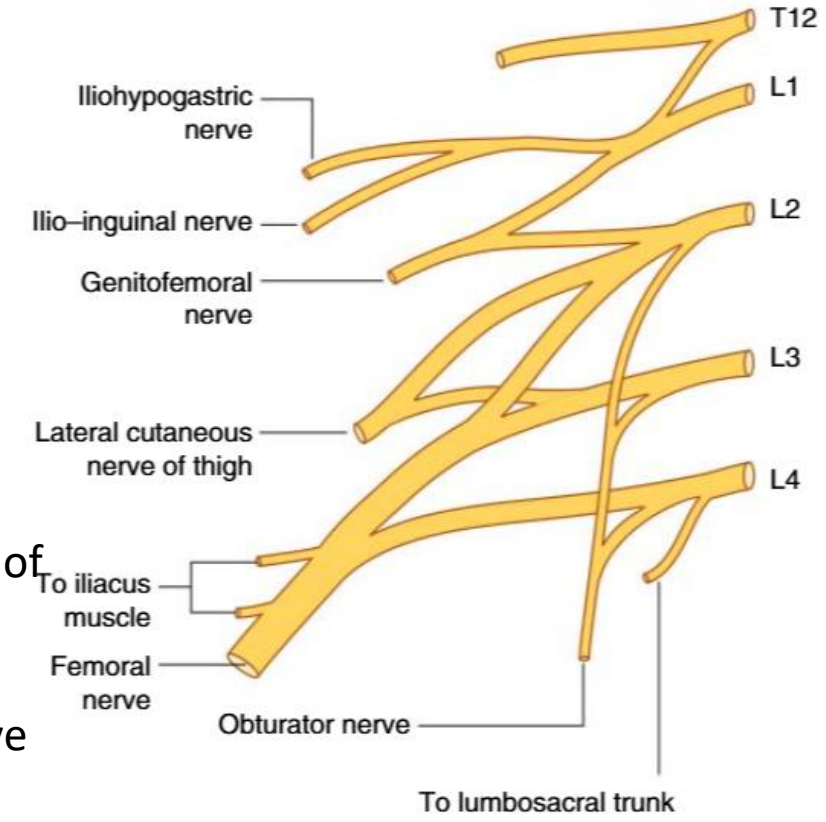
Myotome	Movement
L1 & L2	Hip Flexion
L3 & L4	Knee Extension
L5 to S2	Knee Flexion
S1 & S2	Ankle Plantarflexion
S2 & S3	Adduction of the toes

Lumbar Plexus



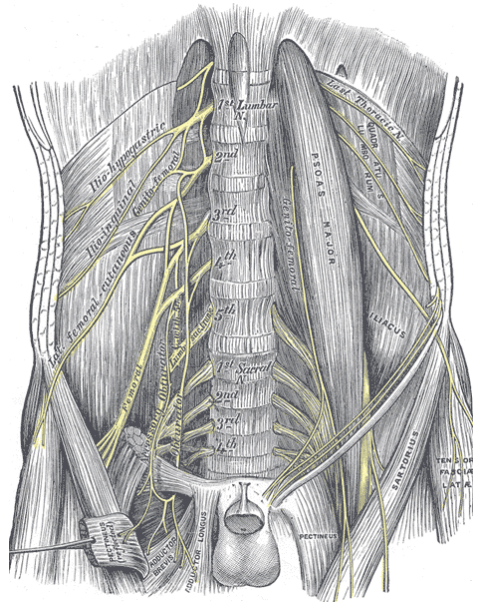
Key outputs

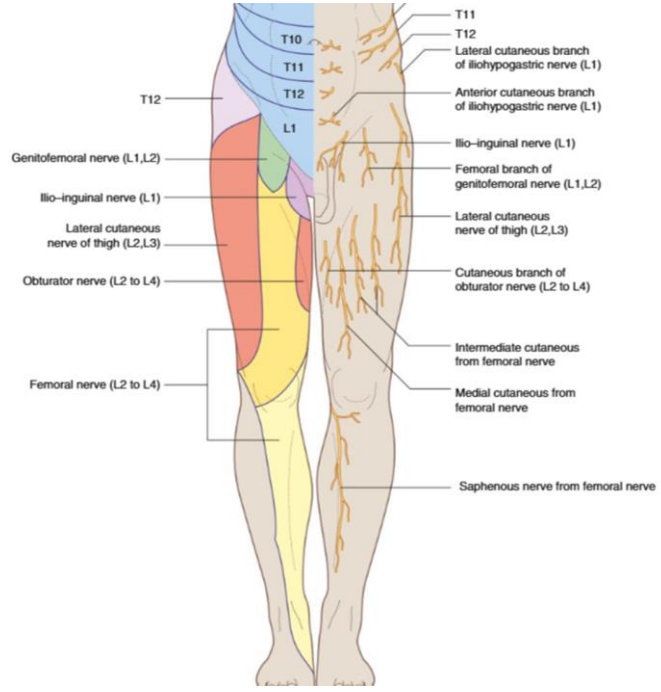
- Subcostal N
- Iliohypogastric N
- Ilioinguinal N
- Lateral cutaneous N of thigh
- Femoral Nerve
- Genitofemoral; Nerve
- Obturator Nerve
- Lumbosacral Trunk



Iliohypogastric Nerve

- **Origin**: anterior ramus of the first lumbar nerve (L1) in the lumbar plexus.
- **Course**: it runs inferiorly and laterally, passing between the quadratus lumborum muscle and the kidney. It then penetrates the transversus abdominis muscle and runs between the transversus abdominis and internal oblique muscles.
- **Branches**: it divides into the anterior and lateral cutaneous branches:
- **Lateral cutaneous branch**: Supplies the skin of the superior gluteal region
- **Anterior branch**: Supplies sensation to the lower abdominal skin above the pubis
- **Sensory distribution**: The iliohypogastric nerve supplies sensation to the lower abdominal skin above the pubis and the skin of the superior gluteal region
- **Damage:** can occur during abdominal procedures, pelvic injuries, sports injuries, or rarely during pregnancy. This can cause sensory deficits or painful paraesthesias.





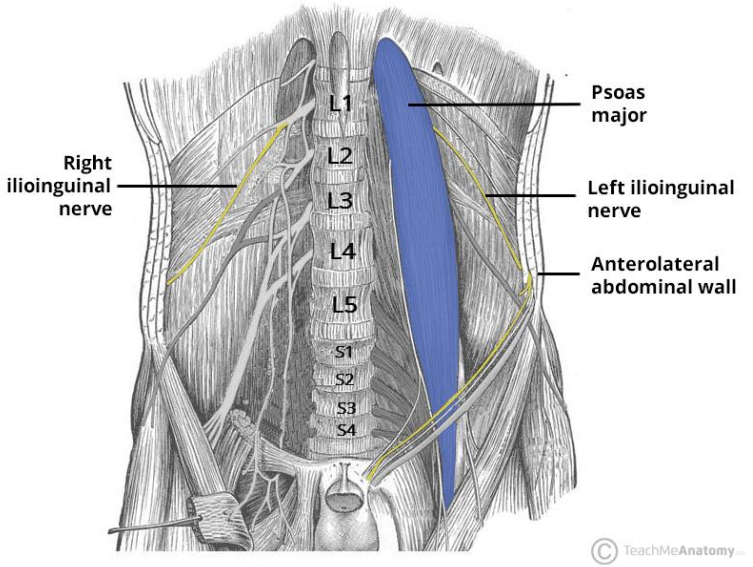
Ilioinguinal Nerve

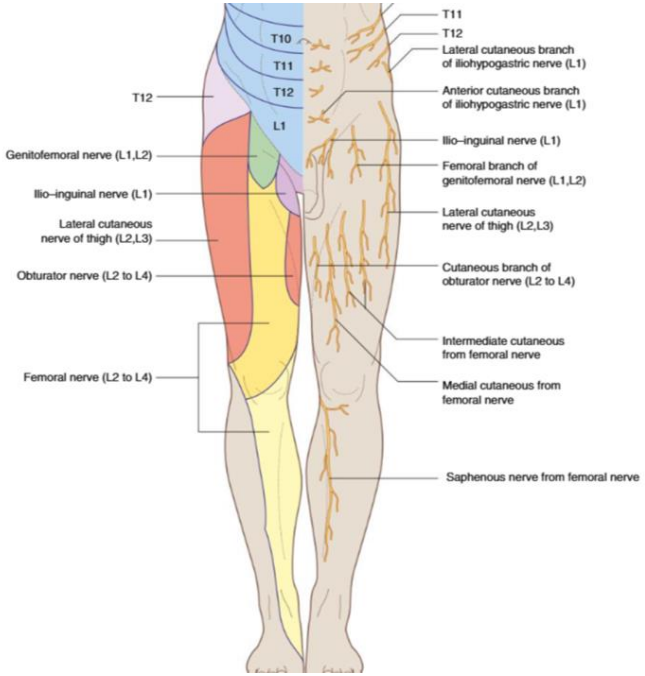
- **Origin:** Anterior ramus of the L1.

- **Course:**
- **Abdominal wall: E**merges from the psoas major muscle, passes in front of the quadratus lumborum, and travels obliquely across the iliacus muscle.
- **Transversus abdominis and internal oblique: P**asses through these muscles.
- **Inguinal canal:** Enters the inguinal canal through the superficial inguinal ring.

- **Innervation:** In males, the nerve innervates the scrotum, while in females, it innervates the labium. It also supplies sensation to the proximal medial thigh.

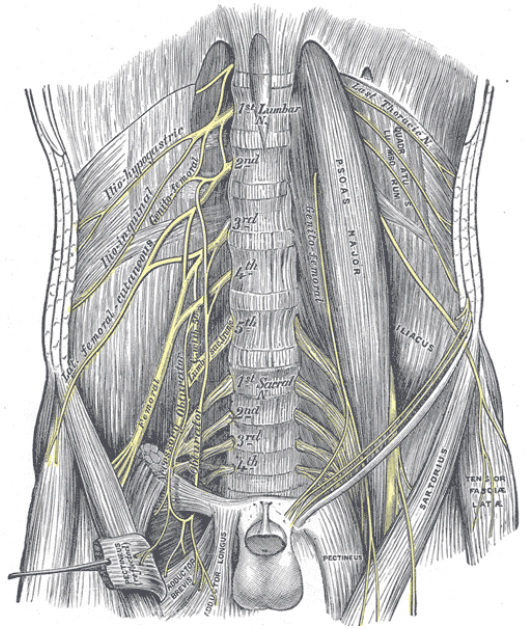
- **Damage:** can cause chronic pain in the groin region, known as post hernia rafi pain syndrome or inguinal neuralgia.
- **Symptoms:** burning pain and numbness in the lower abdomen, genitalia, and inner thigh.

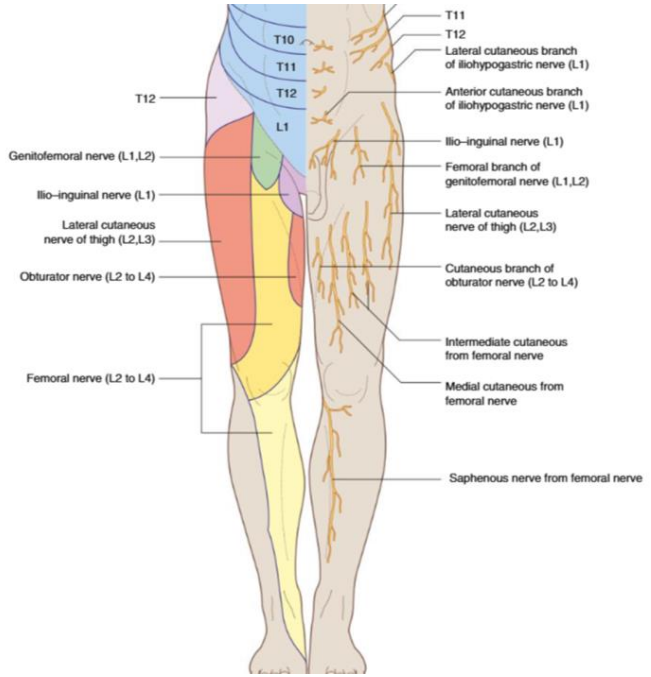




Genitofemoral nerve

- **Origin: A**rises from spinal root L1 & L2
- **Course:** Descend within Psoas major muscle, then pierces it to emerge on the anterior surface.
- **Retroperitoneum**: It descends further in the retroperitoneum, behind the ureter, gonadal vessels, and abdominal vessels.
- **Inguinal ligament:** It divides into genital and femoral branches just above the inguinal ligament.
- The genital branch passes through the deep ring of the inguinal canal. In men: cremasteric muscle; in women skin over labium majus.
- The femoral branch runs under the inguinal ligament and enters the femoral sheath. It then pierces the sheath and fascia latae to become superficial in the anterior thigh. It supplies the skin over the medial surface of upper thigh

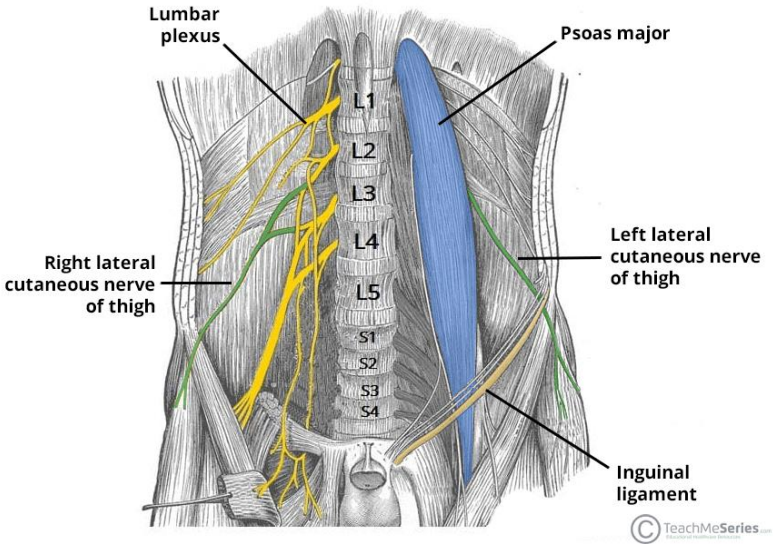


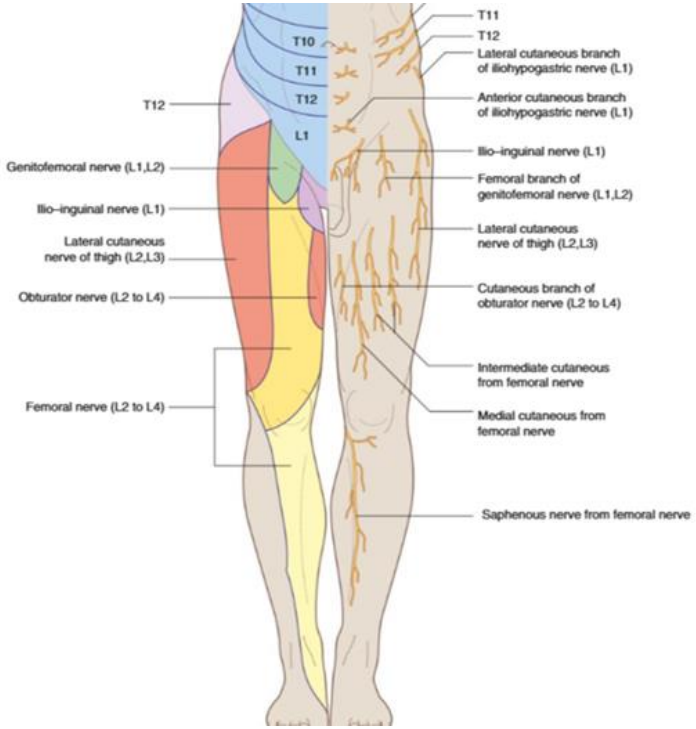


Lateral cutaneous Nerve of thigh

Origin: arises from posterior rami of L2 & L3

Course: emerges from the lateral border of psoas major muscle and passes downwards obliquely, along the anterior surface of iliacus. It enters the lower limb under the inguinal ligament medial to the anterior superior iliac spine. It terminates as numerous sensory branches supplying the anterior and lateral surface of the thigh to the level of the knee





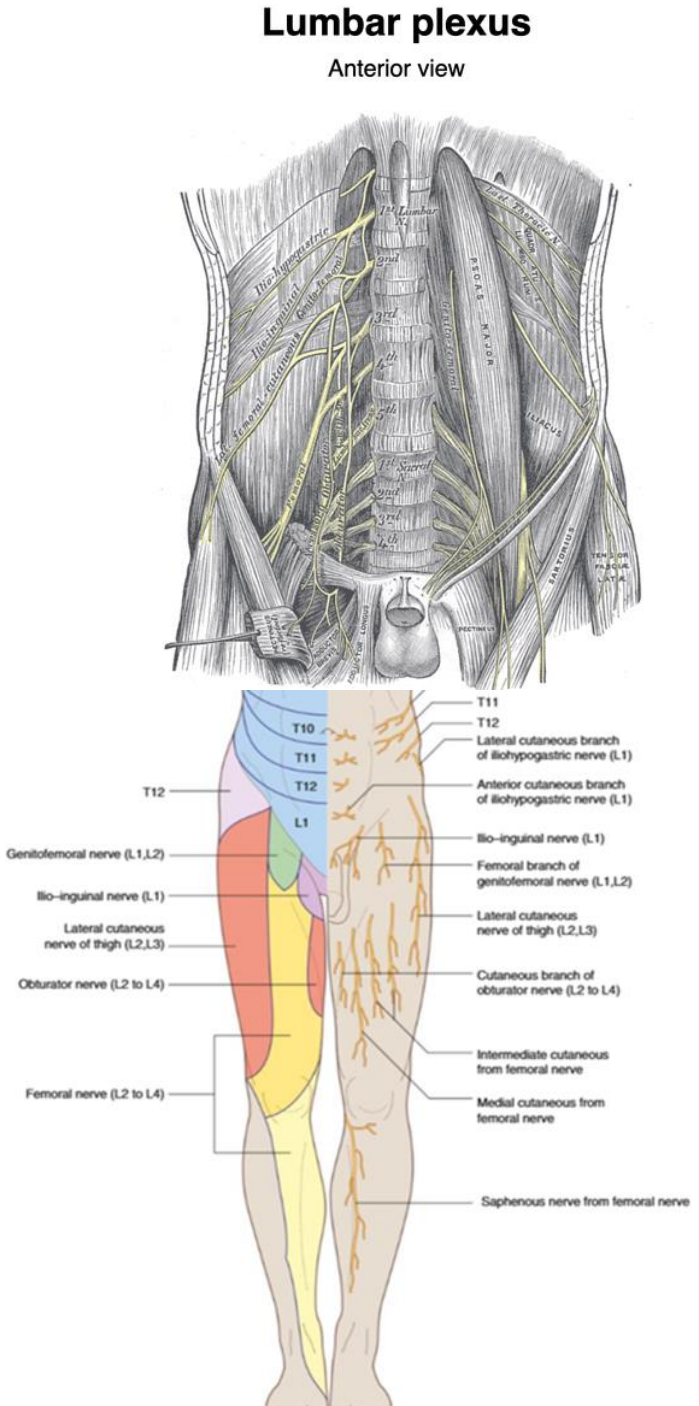
Obturator Nerve (L2-L4)

Origin: anterior rami of L2 – L4

Course: Descends in the psoas major muscle, emerging from its medial side near the pelvic rim. It continues posterior to the common iliac vessels, passes across the lateral wall of the pelvic cavity and enters the obturator canal, through which the obturator nerve gains access to the medial compartment of the thigh. It divides into anterior and posterior branches in the obturator canal. The two branches supply:

Innervations: Articular branches to the hip joint
Muscular branches to the obturator externus, adductor longus, gracilis, adductor brevis, adductor magnus muscles and pectineus
Cutaneous branches to the medial aspect of thigh, and in association with the saphenous nerve, cutaneous branches to the medial aspect of the upper part of the leg and articular branches to the knee joint

There is also an accessory Obturator N in some individuals



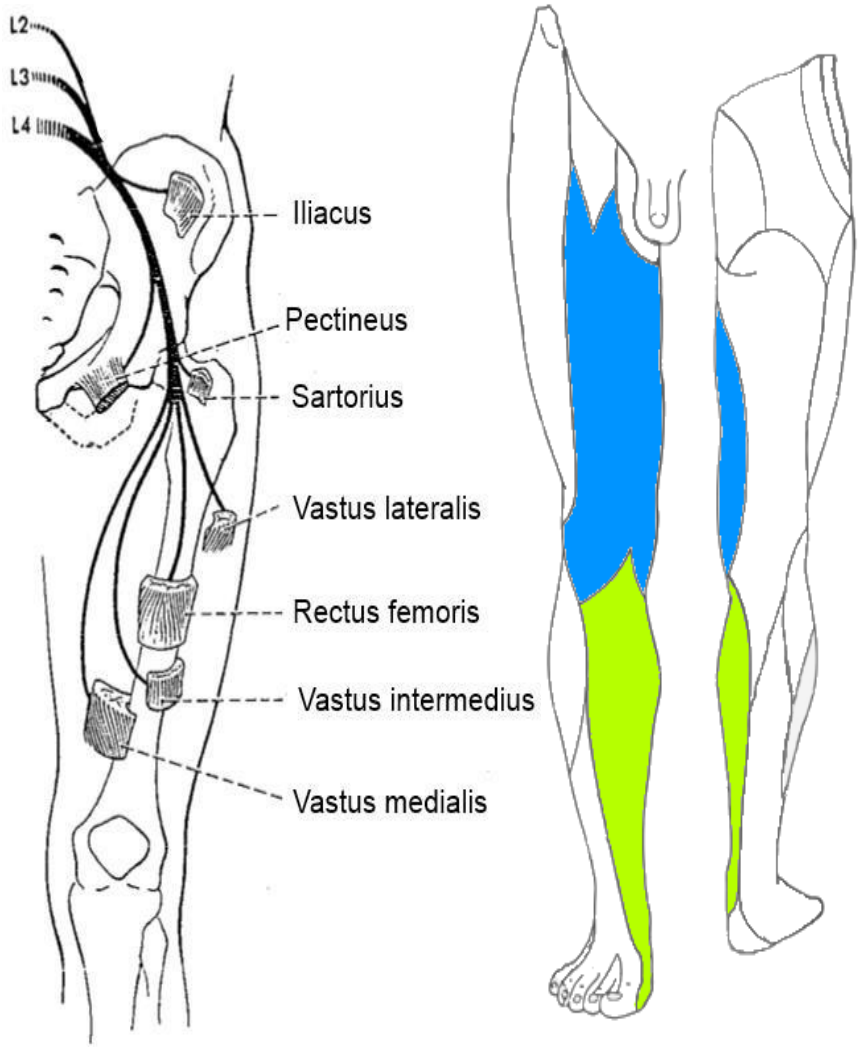
Femoral Nerve (L2-L4)

Origin: Anterior rami of nerve roots L2, l3 & L4

Course: It descends through the substance of psoas major muscle, emerging from the lower lateral border of the muscle. It descends between psoas major and anterior surface of iliacus, deep to the Iliacus fascia and lateral to the femoral artery as it passes posterior to the inguinal ligament and enters the anterior compartment of the thigh. In the thigh it divides into anterior and posterior branches

Innervations: Cutaneous branches: medial an intermediate cutaneous nerves supplying the skin on the anterior surface of thigh & saphenous nerve supplying the skin on the medial surface of the leg.
Muscular branches innervate the iliacus, pectineus, sartorius, rectus femoris, vastus medialis, vastus intermedius, vastus lateralis and articular branches to the hip and knee joints.

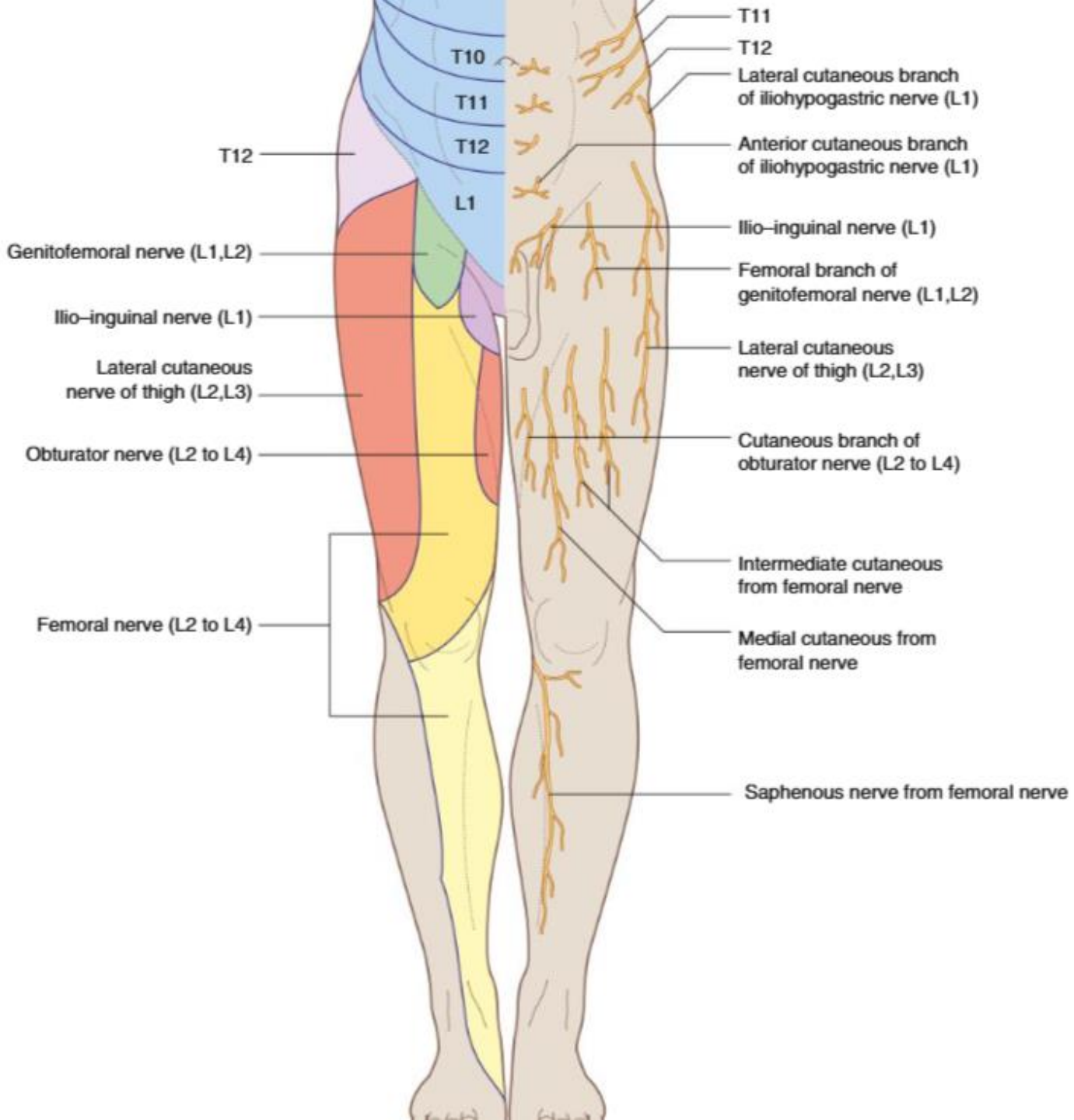
The terminal cutaneous branch of the femoral nerve is the saphenous nerve



Branches of the Lumbar plexus

Branch	Spinal Segment	Function (Motor)	Function (sensory)
Iliohypogastric N	L1	Internal oblique and tranversus abdominis muscle	Posterior gluteal skin and skin of the pubic region
Ilioinguinal N	L1	Internal oblique and tranversus abdominis muscle	Skin in the upper medial thigh, and either the skin over the penis, and anterior scrotum or the mons pubs and labium majus
Genitofemoral N	L1, L2	Genital branch – male cremasteric muscle	Genital branch: Skin of anterior scrotum, or the mons pubs and labium majus; femoral branch – skin of upper anterior thigh.
Lateral cutaneous N of thigh	L2, L3		Skin on anterior and lateral aspect of thigh
Obturator N	L2, L3, L4	Obturator externus, pectineus, muscles of medial compartment of thigh	Skin on medial aspect of thigh
Femoral N	L2, L3, L4	Iliacus, pectineus, muscles of the anterior compartment of thigh	Skin on anterior thigh and medial surface of leg

Cutaneous distribution of the lumbar plexus

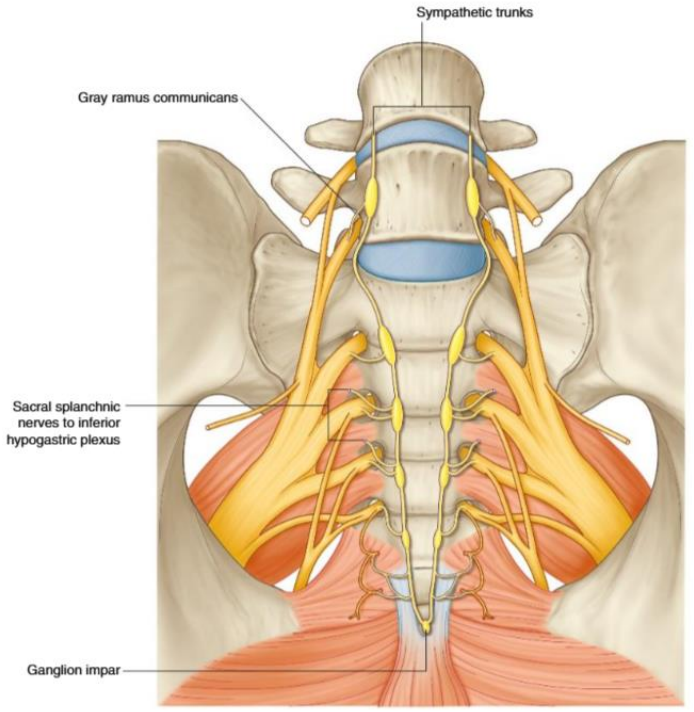


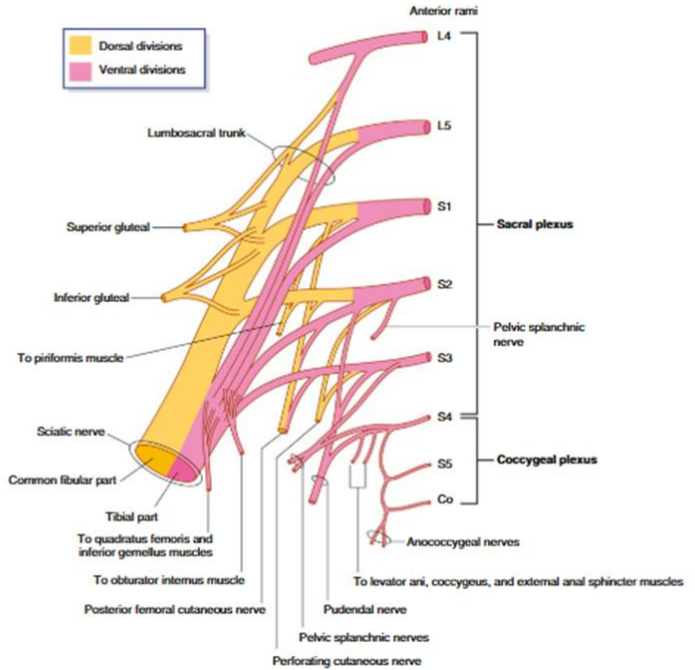


https://youtu.be/vgMC54tmN0M

Sacral Plexus

- Formed by the anterior rami of **S1 to S4** and the lumbosacral trunk **(L4 & L5)**
- Anterior surface of the piriformis muscle in the posterolateral pelvic wall
- Branches of the sacral plexus includes:
  - Sciatic Nerve
  - Pudendal Nerve
  - Gluteal nerves
  - Nerves to the pelvic wall,
  - Pelvic floor and
  - Lower limbs



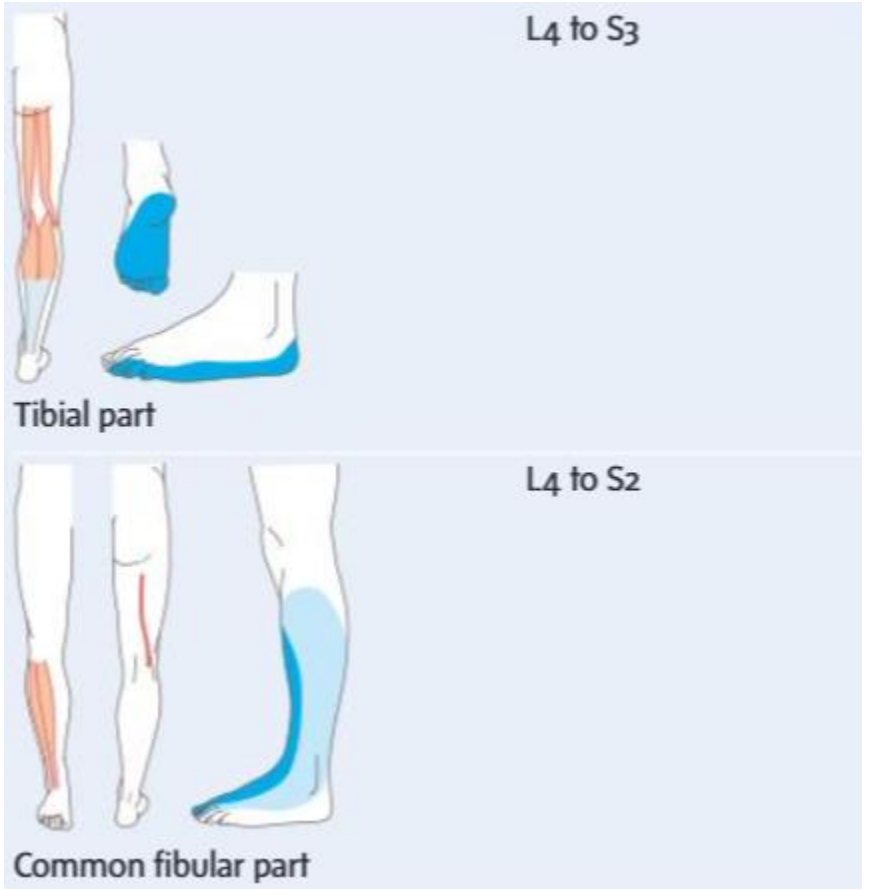


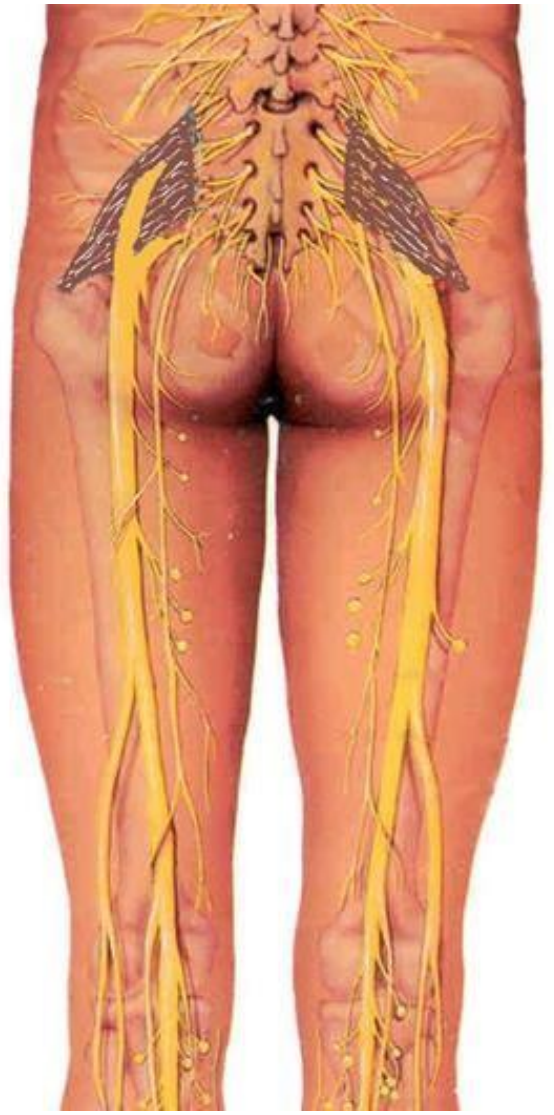
Sciatic Nerve (L4-S3)

Origin: arises from L4 to S3 nerve roots

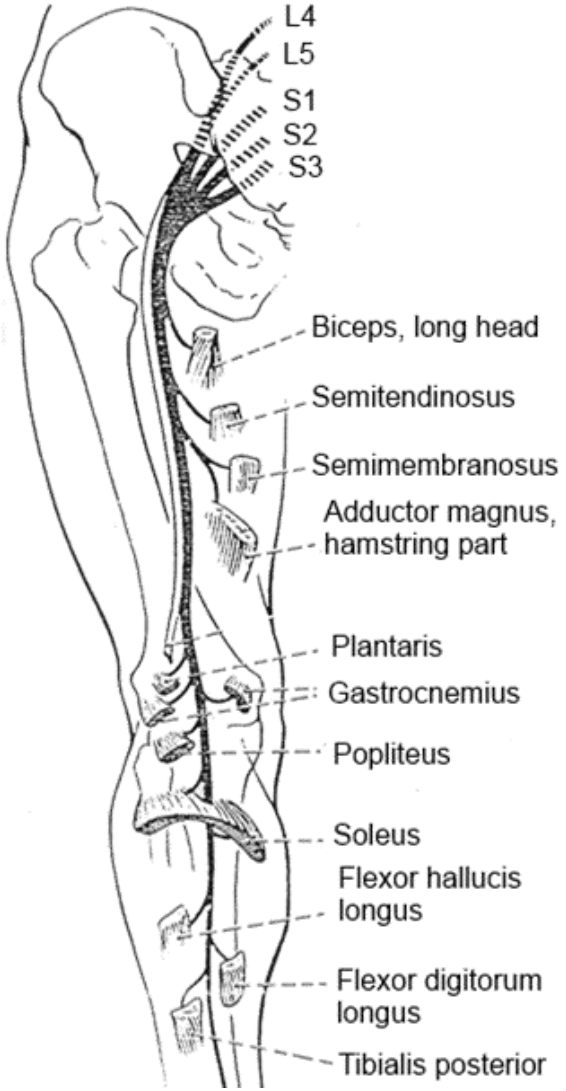
Course: forms on the surface of the piriformis muscle and leaves the pelvic cavity through the greater sciatic foramen inferior to the piriformis. Passes through the gluteal region into the thigh, where it divides into 2 major branches – common fibular (peroneal) nerve (L4, L4, S1 & S2) and the Tibial nerve (L4, L5, S1, S2 & S3)

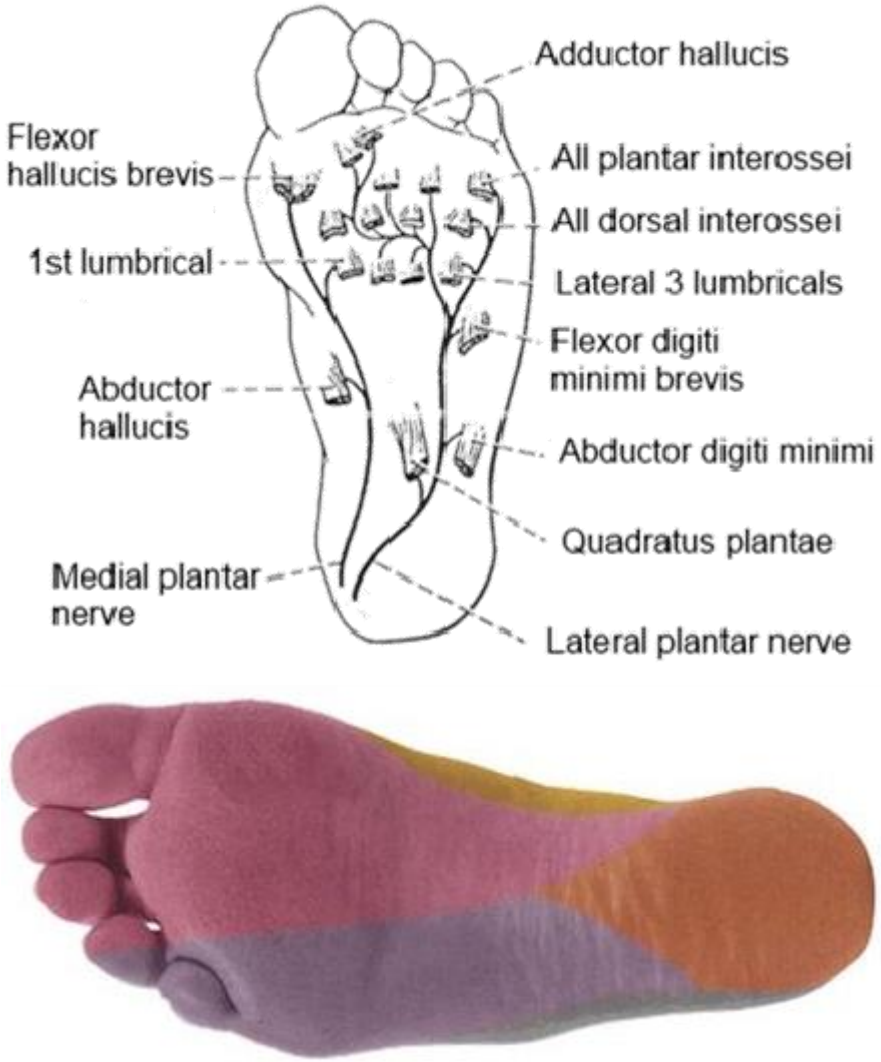
Innervations: muscles in the posterior compartment of thigh (hamstrings), leg and all muscles in the sole of the foot
Cutaneous fibres from skin of the foot and lateral leg

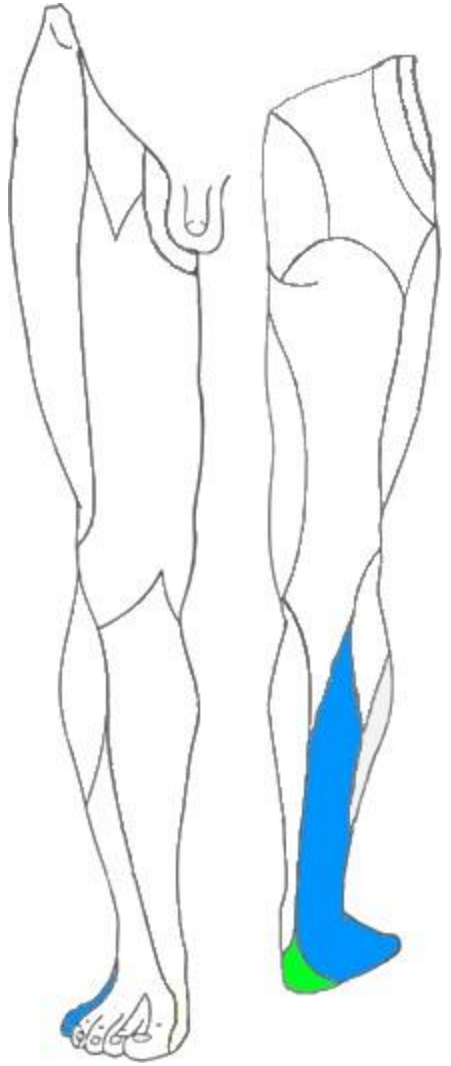




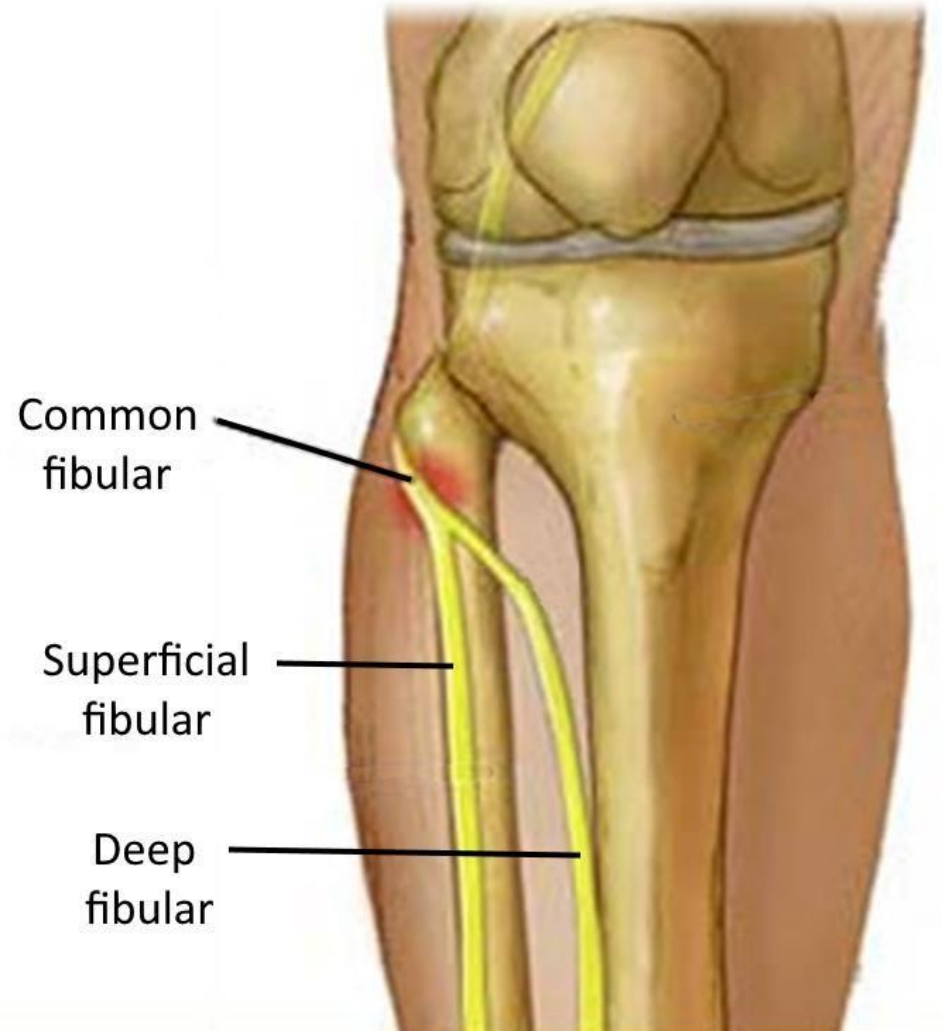
Tibial Nerve (L4-S3)





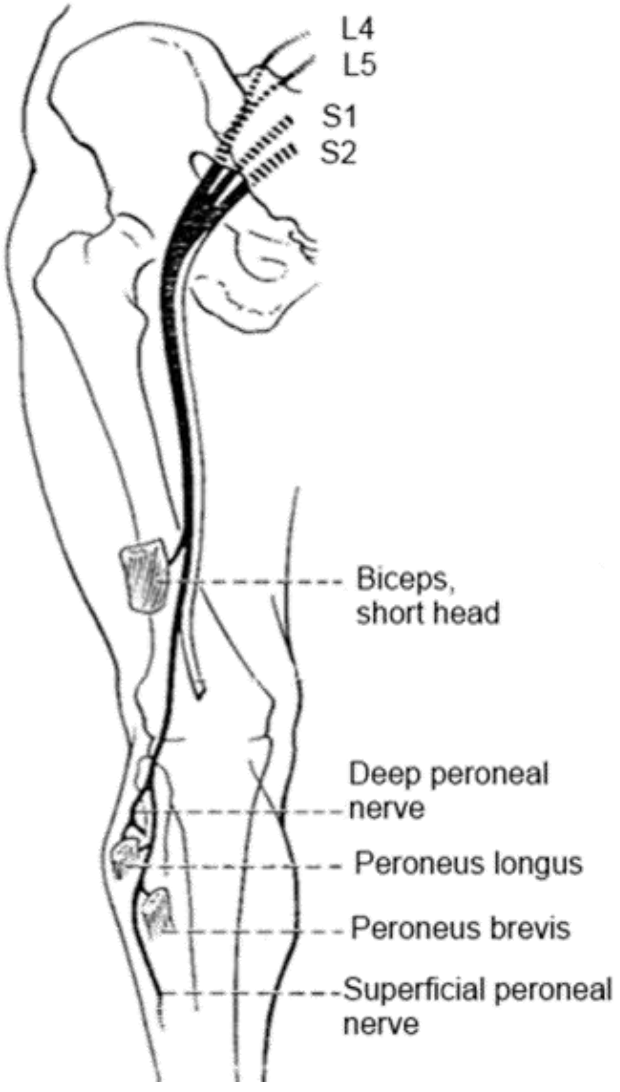


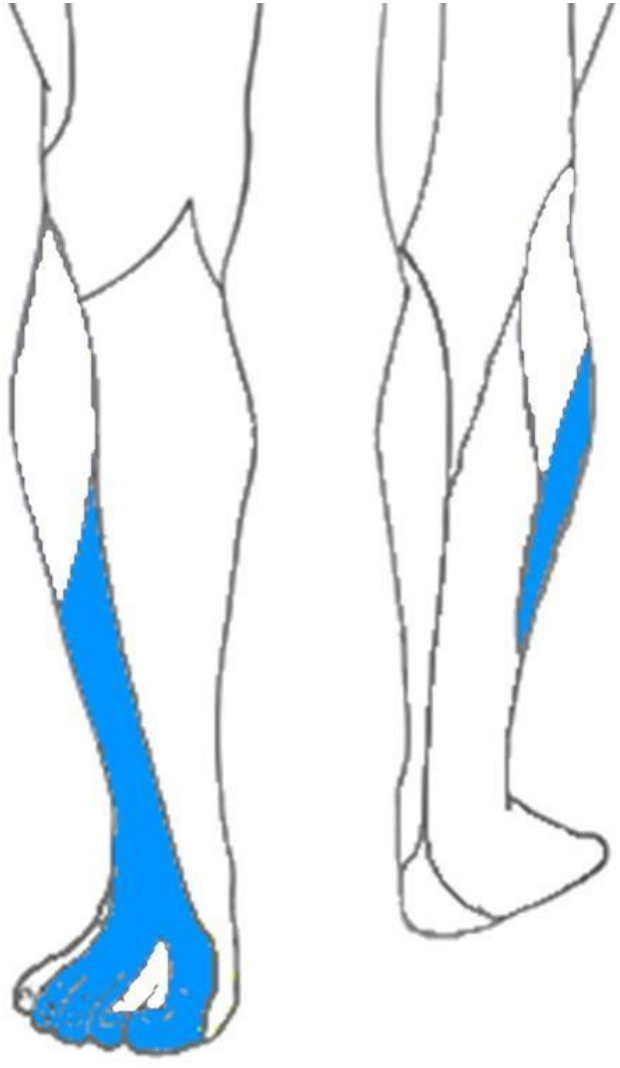
Common Fibular (Peroneal) Nerve (L4-S2)



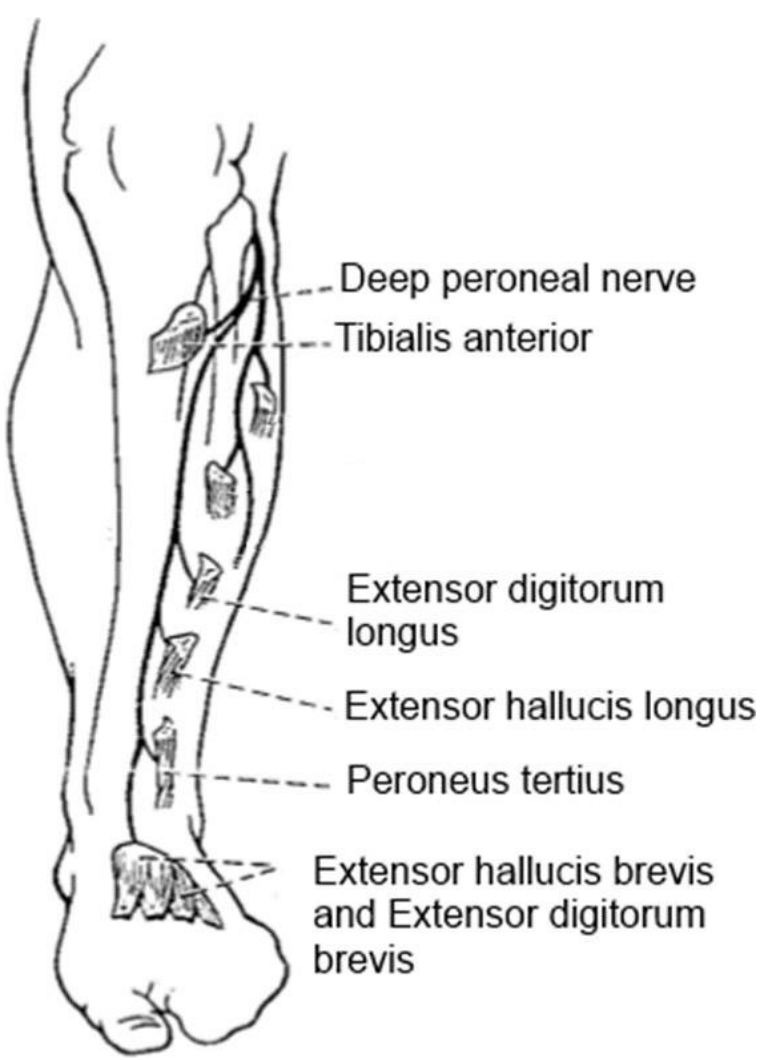


Superficial Fibular Nerve (L4-S2)





Deep Fibular Nerve (L4-S2)





Pudendal Nerve

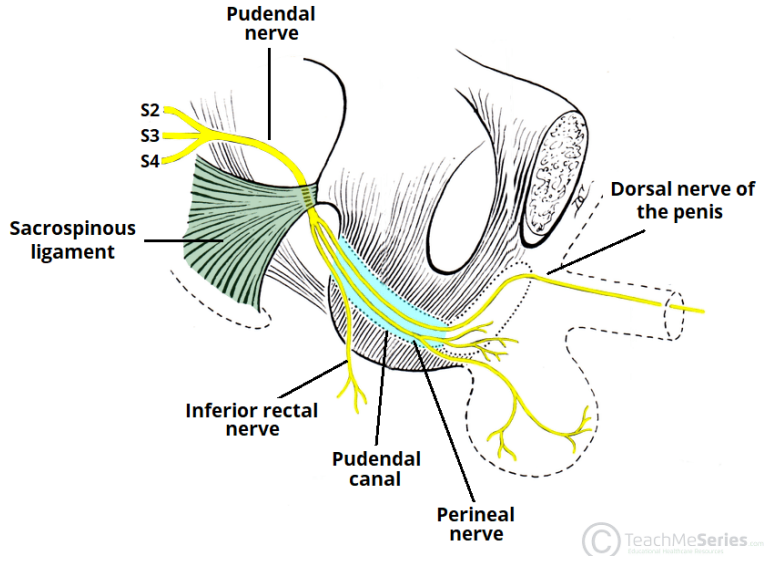
Origin: Originate from the anterior rami of S2 to S4

Course: It leaves the pelvic cavity through the greater sciatic foramen inferior to piriformis, muscle and enters the gluteal region. It courses into the perineum through the lesser sciatic foramen and traves through the pudendal canal. It divides into branches : inferior rectal nerve, perineal nerve and continues and as the dorsal nerve of penis or clitoris

Innervations: Innervates skin and skeletal muscles in the perineum, including the external anal and external urethral sphincters

Clinical relevance:

- Pudendal block anaesthesia.
- Pudendal nerve stimulation
- Pudendal neuralgia

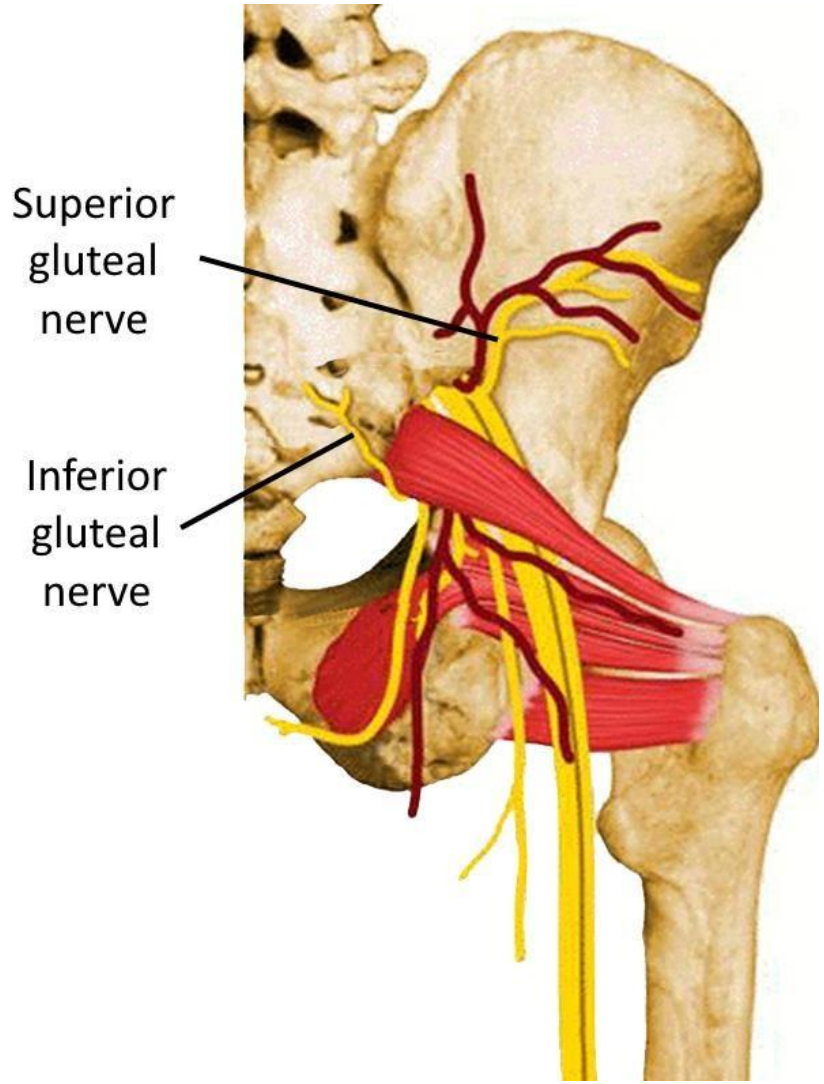


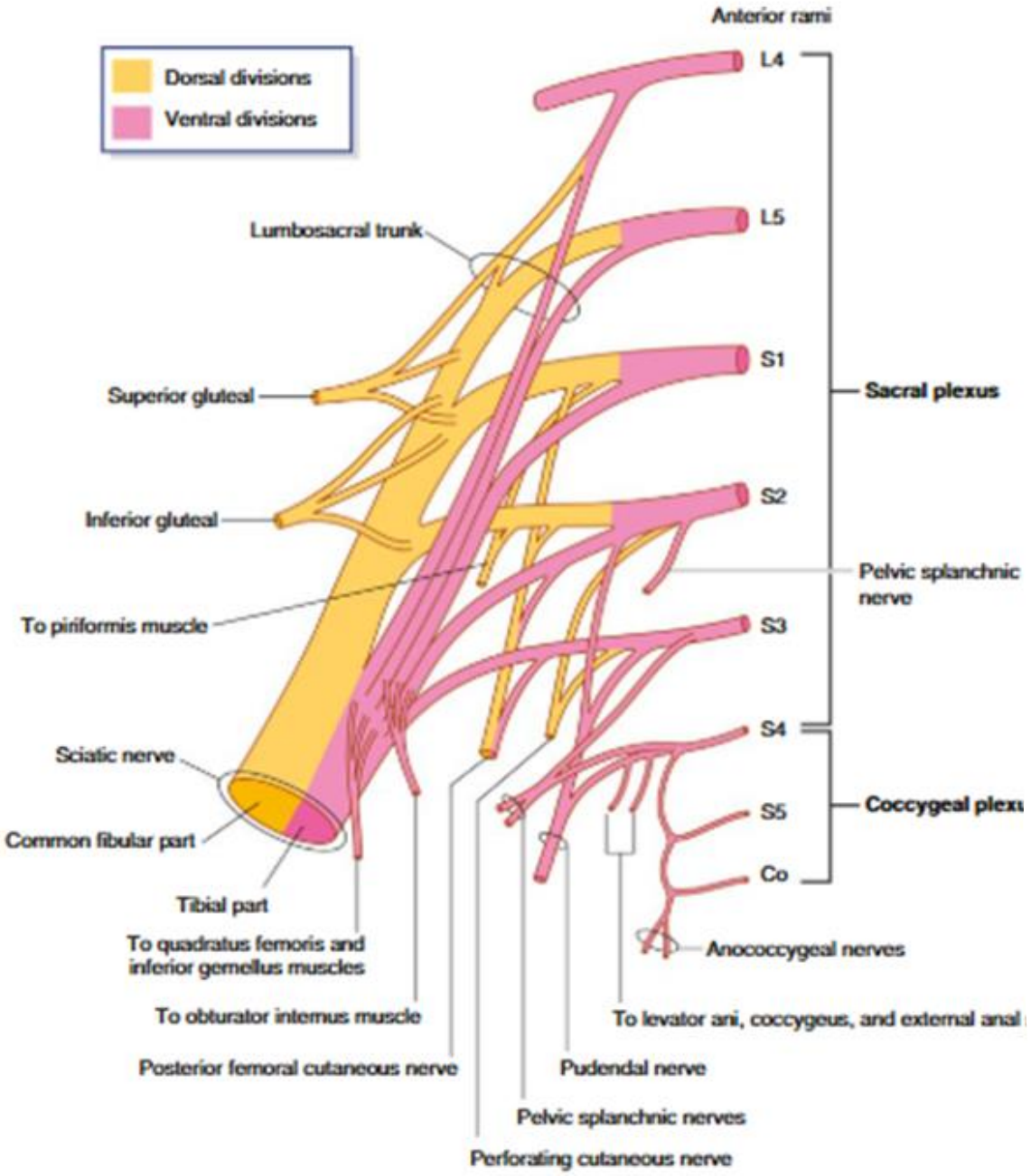
- Inferior rectal nerve: external anal sphincter
- Perineal nerve: muscles of the pelvic floor and perineum
  - bulbospongiosus
  - Ischiocavernosus
  - Levator ani muscles
  - External urethral sphincter
- Dorsal nerve of penis or clitoris

Gluteal Nerves (L4-S2)

- Superior gluteal (L4-S1)
  - » Gluteus medius
  - » Gluteus minimus
  - » Tensor fasciae latae
  - » Hip joint

- Inferior gluteal (L5-S2)
  - » Gluteus maximus

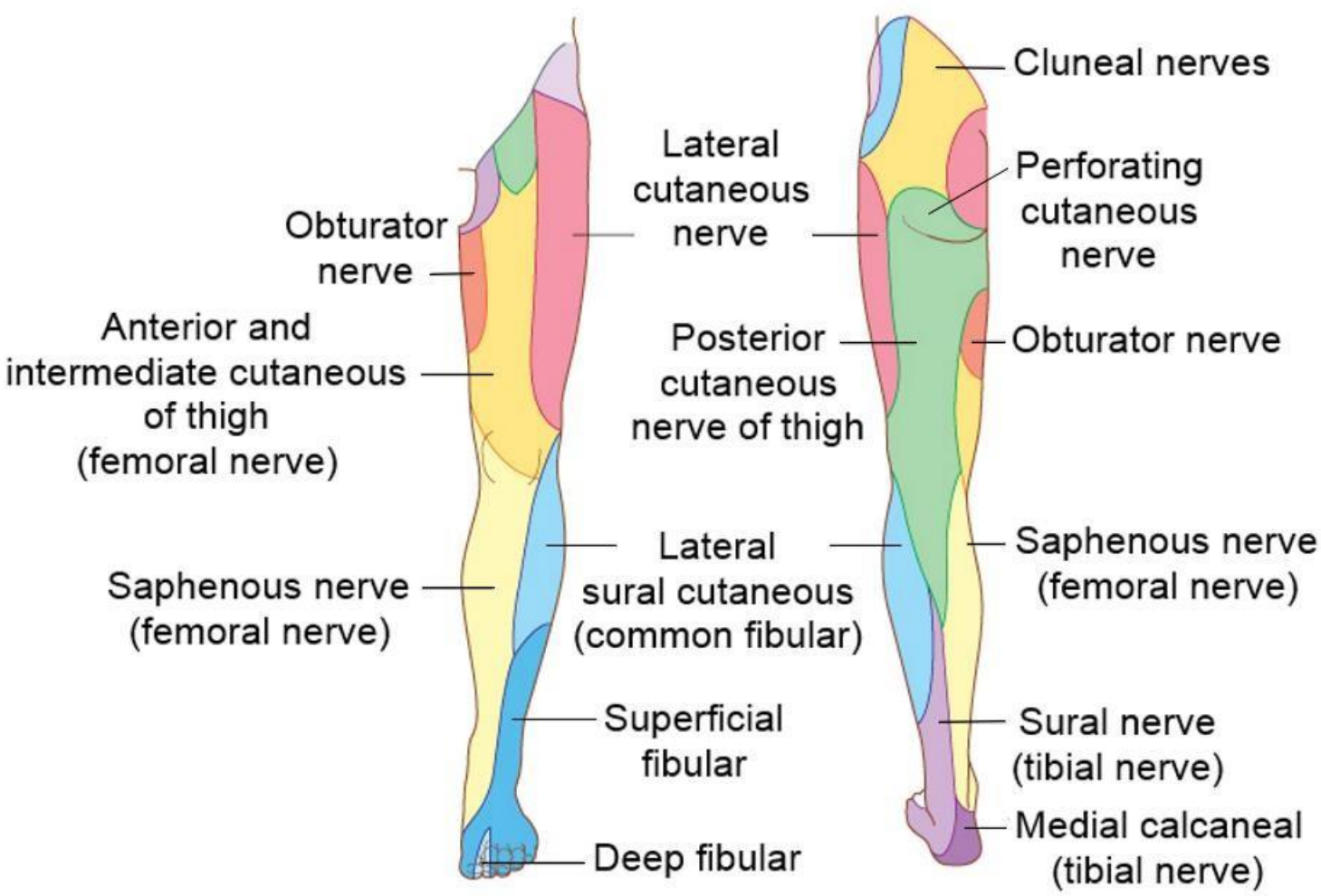


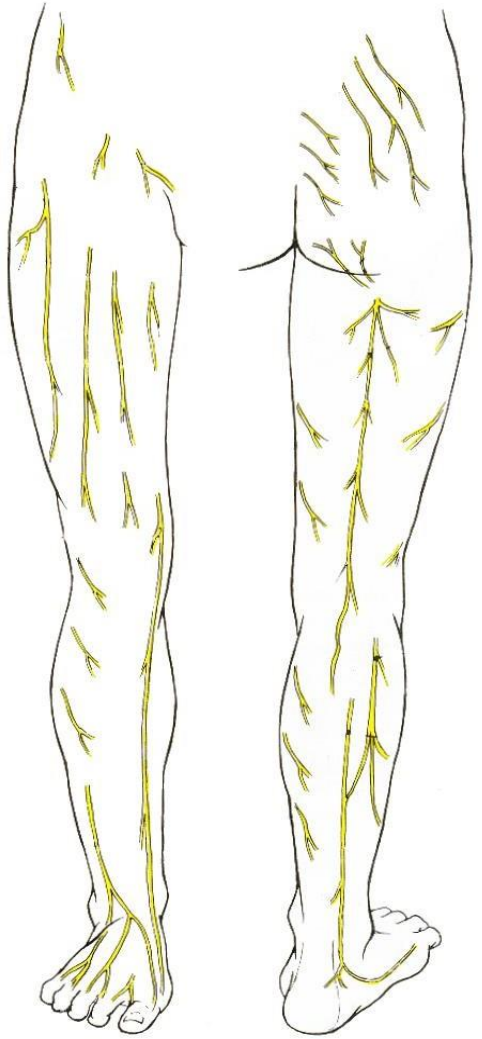


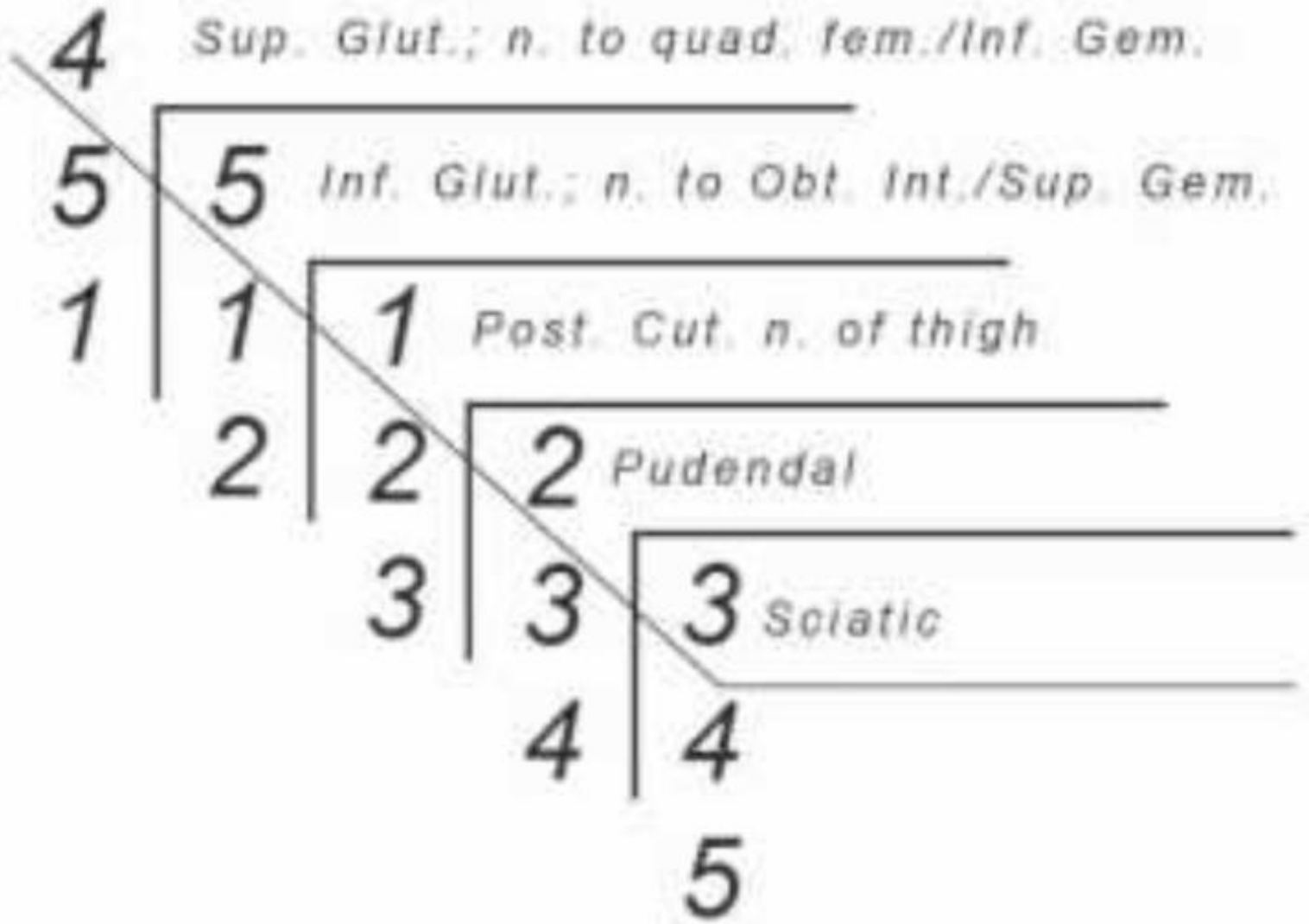
Other branches of the Sacral Plexus

- Nerve to Obturator Internus & associated superior gemellus muscle (L5 to S2)
- Nerve to quadratus femoris and the inferior gemellus muscle (L4 to S1)
- Posterior femoral cutaneous nerve (S1 to S3)
- Perforating cutaneous nerve (S2, S3)
- Nerve to the piriformis (S1 & S2)

Cutaneous Nerves of the Lower Limb







https://youtu.be/w3t8haJCbAQ

What must I know

1. What is a Dermatome?
2. What is a Myotome?
3. Myotomal distribution of the lower limb joint movements
4. Dermatomal map of the lower limb
5. Name and roots of lower Limb nerves
6. Course and distribution of common lower limb nerves
7. Cutaneous nerve distribution of the lower limb

